

Day 4

Integration &

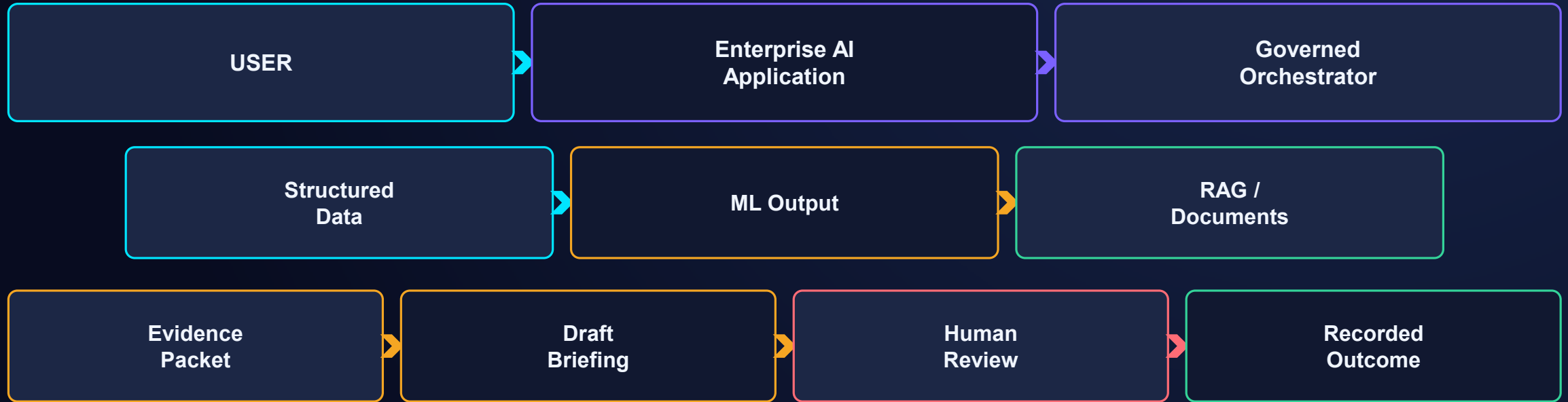
the A-001 Team Capstone

One inspectable workflow - then teams prove control, not just intelligence.

- Morning: integrate data, ML, RAG and the agent into one workflow
- Keep evidence types separate, labelled and traceable
- Afternoon: team capstone - an evidence-backed review briefing
- Every run ends at a recorded human decision

One Inspectable Workflow

Everything from three days, composed - nothing rebuilt.



Inherited, not rebuilt: Day 1 governed data + forecast · Day 2 cited RAG · Day 3 bounded tools + skills + state

Audit events are written at every transition - not reconstructed afterwards.

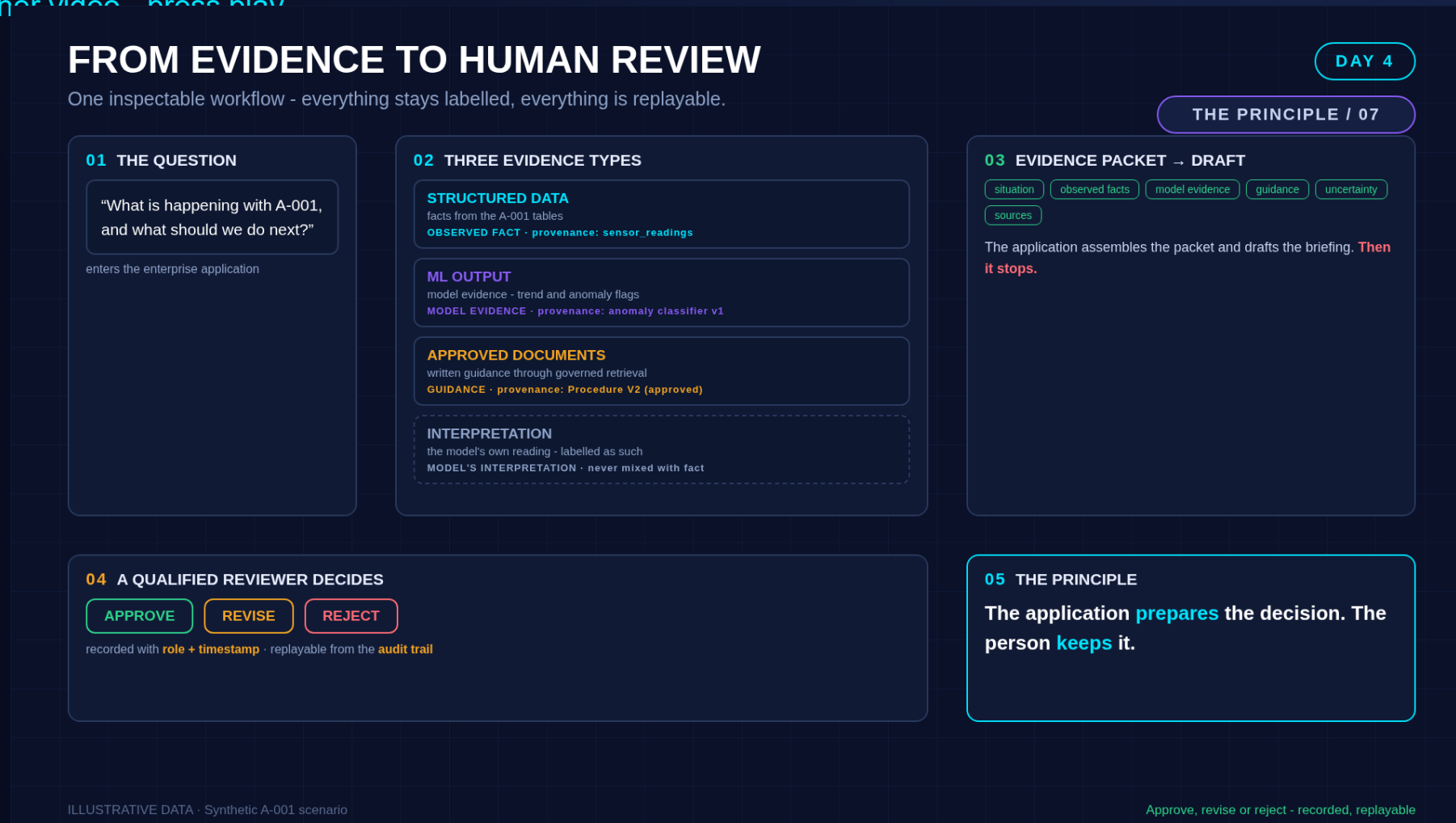
Watch: The Whole Machine, End to End

Narrated explainer video - press play



Watch: From Evidence to Human Review

Narrated explainer video - press play



What Must Stay Separate

Five evidence types - five different kinds of trust.

Evidence type	Example	Why separation matters
Observed structured facts	Health score, work-order count	Directly queryable facts
Sensor / model evidence	Trend, forecast, MAE	Analytical evidence - not a diagnosis
Written guidance	Approved V2 procedure	Organisational authority
Model interpretation	Summary or recommendation	Generated reasoning - must be traceable
Human decision	Approve / revise / reject	Final authority

The moment a briefing blurs a forecast with a fact, or a draft with an approval, review becomes impossible.

Watch: The Audit Trail - Replaying Any Decision

Narrated explainer video - press play

THE AUDIT TRAIL

Replaying any decision - the difference between trust and verification.

01 · THE RUN RECORD

the request · the guard's verdict · which tool was chosen, and why

02 · EVERY TOOL CALL

the query · rows or chunks returned · timestamps · nothing the agent saw is invisible

03 · THE EVIDENCE PACKET, STORED WHOLE

model scores · cited documents · versions and statuses · exactly as presented

04 · THE HUMAN DECISION

the draft · who reviewed, when · approve, revise or reject · with their comment

if you cannot replay it, you did not govern it

trust vs verification

ILLUSTRATIVE DATA · Synthetic A-001 scenario

DAY 4

SENSOR TO SIGNATURE / 06

REPLAY = RECONSTRUCTION

From the trail alone we rebuild **what the agent knew**, **what it wrote**, and **where the human intervened**.

For A-001: any work order, cited procedure or flagged reading traces end to end - sensor to signature.

Replay, or it was not governed

The Capstone Scenario

A-001 Condition / Reliability Review - decision support, end to end.

CAPSTONE QUESTION

"What does the available evidence say about A-001, what should a qualified reviewer pay attention to, and what is the appropriate human next step?"

REQUIRED EVIDENCE

- Structured enterprise data
- Recent sensor trend and/or ML output
- Approved document guidance
- Work-order / inspection / field context
- Provenance and source identifiers
- Evidence gaps, conflicts, uncertainty

THE BOUNDARY - DECISION SUPPORT

The application may prepare evidence and a review briefing. It must not diagnose a real failure, issue control commands, authorise maintenance, close work orders, treat a model output as approval, or approve its own recommendation.

3.5 days of capability - one afternoon to prove you can govern it.

Watch: The Capstone - Prove Control, Not Just Intelligence

Narrated explainer video - press play

PROVE CONTROL, NOT JUST INTELLIGENCE

The capstone: decision support for A-001, end to end.

DAY 4

IN CONTROL / 07

01 THE CAPSTONE QUESTION

"What does the available evidence say about A-001, what should a qualified reviewer pay attention to, and what is the appropriate human next step?"

02 THE FULL STACK

Structured data
the governed A-001 tables

Sensor trend + anomaly flags
the ML output, labelled as model evidence

Approved documents
through governed retrieval

The agent's tool path
every call inspectable
→ one bounded briefing

03 THE BOUNDARIES ARE THE POINT

- ✗ No diagnosing a real failure
- ✗ No control commands
- ✗ No authorising maintenance
- ✗ No approving its own recommendation

04 A STRONG DEMO SHOWS SIX THINGS

1 the business question

2 the evidence and its sources

3 what the application did

4 what it found

5 what it cannot conclude

6 where the human remains responsible

05 THE ARC

Gathers evidence, preserves uncertainty -
keeps a person in control.

We started with a table of sensor readings. We end here.

ILLUSTRATIVE DATA · Synthetic A-001 scenario

Control counts as much as intelligence

Ten Steps, One Boundary

The capstone workflow - evidence first, briefing second, human last.

01 Restate the business question

One sentence, agreed by the team

02 Inspect the structured data

What do the tables show?

03 Review sensor trend + ML output

What does the forecast suggest - and not prove?

04 Retrieve relevant documents

Governed retrieval, not naive search

05 Confirm the current authority

Which procedure version governs?

06 Combine without blurring

Keep evidence types labelled

07 Name conflicts and gaps

What is unknown or unverified?

08 Draft the bounded briefing

The 8-section template

09 Stop at the human boundary

Reviewer approves, revises or rejects

10 Prepare the demonstration

Six beats: problem → human next step

Teams and Roles

Small teams, clear ownership - roles can combine.

Role	Primary responsibility
Team facilitator	Keep the team on the business question; timebox decisions
Data lead	Structured-data exploration and ML evidence
Knowledge / RAG lead	Document authority, retrieval and citations
Agent / integration lead	Tool routing, evidence packet, workflow controls
Reviewer / presenter	Challenge assumptions, check limits, lead the demo

INSTRUCTOR GUIDANCE

Form teams before the final half-day where possible. During the capstone, do not solve the project for them - ask evidence and boundary questions instead: "Which version governs? What is the source of that number? Where does the human decide?"

The Eight-Section Briefing

Decision support a reviewer can actually work with.

1 · Situation

What triggered the review?

3 · Predictive evidence

ML output - and what it does NOT prove

5 · Assessment

What the combined evidence supports

7 · Human next step

What a qualified reviewer decides next

2 · Structured evidence

Database, work and sensor records

4 · Document evidence

Approved docs: IDs, versions, status

6 · Gaps / uncertainty

Unknown, conflicting, unverified

8 · Sources used

Tables, views and document IDs

Sections 3, 6 and 7 carry the rubric: limits, honesty about gaps, and a clear human next step.

The Evaluation Rubric

Five dimensions, five points each - control counts as much as intelligence.

Dimension	What good work demonstrates
Problem understanding	Frames the business question; no leap to a technical answer
Data use	Structured data used appropriately; observation vs interpretation separated
AI / RAG use	Relevant knowledge retrieved; approval and version status respected
Evidence & explainability	Claims traceable to sources; limitations identified
Governance & human oversight	Uncertainty preserved; final authority stays with a human reviewer

21-25 strong · 16-20 good · 11-15 partial · 5-10 guided remediation

Training feedback, not an operational competency certification.

The Final Demonstration

Six beats - and the questions your instructor will ask.

THE SIX-BEAT DEMO STORY

1. Problem - the business question in one sentence
2. Evidence - what data and documents were used
3. What the AI application did - the tool path
4. What it found - with source IDs
5. What it cannot conclude - limits and gaps
6. What a qualified human reviews next

QUESTIONS TO EXPECT

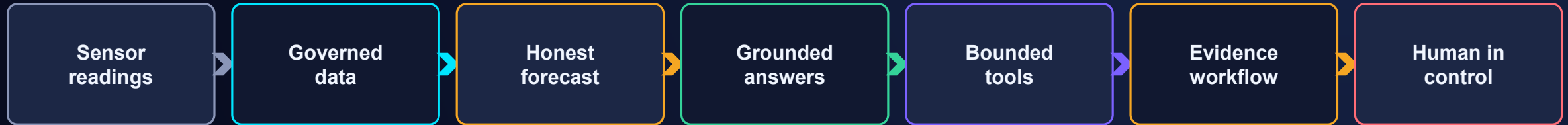
Why did you trust this document, not the other version?
Which part came from data, which from written guidance?
What if retrieval returned no approved evidence?
Why is your agent allowed to call this tool?
What decision is deliberately left to a person?
What would you improve before production?

A fluent demo that cannot answer these six questions has not finished the course.

CLOSE

From a Sensor Table to a Governed Application

The whole four days, one arc.



THE CLOSING MESSAGE

We started with a table of sensor readings. We ended with an AI-enabled application that can understand structured data, make a prediction, retrieve trusted organisational knowledge, select bounded tools, assemble evidence, preserve uncertainty - and keep a human reviewer in control.

What to take back to work: pick ONE real workflow, apply the same arc - data, prediction, knowledge, tools, boundary - and write its five questions down before writing any code.

Thank you - keep the evidence trail, keep the human boundary.